

Graphics Card Market Analysis - Web Scraping Project

Project Overview

In this project, you will create a web scraper to collect and analyze graphics card data from Newegg.com's GPU section. You'll use Python, BeautifulSoup, and other relevant libraries to extract key information about graphics cards currently available in the market.

Learning Objectives

- Implement web scraping techniques using Python and BeautifulSoup
- Handle pagination in web scraping
- Process and structure collected data
- Practice working with real-world data
- Learn to respect website scraping policies and implement delays

Part 1: Web scrapping

Required Data to Extract

For each graphics card listing, you must collect:

1. Product Image URL
2. Product Title (e.g., "MSI Ventus GeForce RTX 3060 12GB")
3. Product Details:
 - Manufacturer
 - Model Number
 - Rating (number of stars)
4. Pricing Information:
 - Current Price
 - Original Price (if available)
 - Shipping Information

 Screenshot

Project Requirements

Technical Requirements

1. Use Python 3.x
2. Required libraries:
 - requests

- BeautifulSoup4
- pandas
- time (for implementing delays)

Functional Requirements

1. Implement proper error handling
2. Include appropriate delays between requests (minimum 2 seconds)
3. Save the scraped data to a CSV file with appropriate column headers
4. Document your code with comments
5. Create a summary of the collected data (e.g., average price, price range)

Part 2: Data Analysis

Required Analysis

1. Price Analysis
 - Calculate average price by brand
 - Identify price ranges and market segments
 - Analyze price-to-memory ratio
 - Compare prices across different GPU series (RTX 3000 vs 4000)
 - Create price distribution visualizations
2. Brand Analysis
 - Market share by brand
 - Average ratings by brand
 - Price comparison across brands
 - Create brand comparison visualizations
3. Memory Configuration Analysis
 - Distribution of memory sizes
 - Price correlation with memory size
 - Popular memory configurations by brand
4. Market Segmentation
 - Entry-level vs Mid-range vs High-end segments
 - Price brackets and their features

Deliverables

1. Python script containing the web scraper
2. CSV file with the collected data
3. README file explaining:
 - Project Description
 - Features
4. Short report analyzing the collected data

Important Notes

- Always check and respect the website's robots.txt file
- Implement appropriate delays between requests to avoid overwhelming the server
- Handle your requests responsibly and ethically
- Be prepared to handle cases where data might be missing or in an unexpected format

```
In [11]: import selenium
        from selenium import webdriver
```

```
In [25]: import selenium
        from selenium import webdriver
        # Import chrome webdriver to access chrome options
        from webdriver_manager.chrome import ChromeDriverManager
        from selenium.webdriver.chrome.options import Options
        from selenium.webdriver.chrome.service import Service
        from selenium.webdriver.common.by import By
        from selenium.webdriver.support.ui import WebDriverWait, Select
        from selenium.webdriver.support import expected_conditions as EC
        import pandas as pd
        import time
```

```
In [26]: from tqdm import tqdm
```

```
In [27]: import requests

url2 = "https://www.newegg.com/GPUs-Video-Graphics-Cards/SubCategory/ID-48"
requests.get(url2)
```

```
Out[27]: <Response [200]>
```

```
In [28]: url = "https://www.newegg.com/Desktop-Graphics-Cards/SubCategory/ID-48"
        service = Service(ChromeDriverManager().install())
        options = Options()
        options.add_argument('--headless=new')
        driver = webdriver.Chrome(service=service, options=options)

        gpus = []
        PAGES_TO_SCRAPE = 20

        for page in range(1, PAGES_TO_SCRAPE + 1):
            products = []
            for attempt in range(3):
                driver.get(f"{url}/Page-{page}")
                time.sleep(2)
                products = driver.find_elements(By.CLASS_NAME, 'item-cell')
                if len(products) >= 36:
                    break
                print(f"Page {page} returned {len(products)} products, retrying...")
                time.sleep(3)

            if len(products) < 36:
                print(f"Page {page} failed after 3 attempts, skipping.")
                continue
```

```

for product in tqdm(products, f'Page {page} in progress...'):
    img = product.find_elements(By.CLASS_NAME, 'item-img')
    img = img[0].find_element(By.TAG_NAME, 'img').get_attribute('src') if im

    rating = product.find_elements(By.CLASS_NAME, 'item-rating')
    rating = rating[0].get_attribute('title').split('+ ', 1)[-1] if rating e

    reviews = product.find_elements(By.CLASS_NAME, 'item-rating-num')
    reviews = reviews[0].text if reviews else None

    brand = product.find_elements(By.CLASS_NAME, 'item-brand')
    brand = brand[0].find_element(By.TAG_NAME, 'img').get_attribute('title')

    title = product.find_elements(By.CLASS_NAME, 'item-title')
    title = title[0].text if title else None
    if not title:
        continue
    model = product.find_elements(By.CLASS_NAME, 'item-features')
    model_text = model[0].text.split(':', 1)[-1] if model else None

    price = product.find_elements(By.CLASS_NAME, 'price-current')
    price = price[0].text.split(' ', 1)[0] if price else None

    shipping_price = product.find_elements(By.CLASS_NAME, 'price-ship')
    shipping_price = shipping_price[0].text if shipping_price else None

    original_price = product.find_elements(By.CLASS_NAME, 'price-was')
    original_price = original_price[0].text if original_price else None

    gpus.append({
        'img': img,
        'rating': rating,
        'reviews': reviews,
        'brand': brand,
        'title': title,
        'model': model_text,
        'price': price,
        'shipping_price': shipping_price,
        'original_price': original_price
    })

time.sleep(3)
driver.quit()

df = pd.DataFrame(gpus)
df.to_csv('newegg.csv', index=False)

```

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| 27/36 [00:11<00:04, 2.07it/s]
Page 5 in progress...: 78%|██████████|
| 28/36 [00:12<00:03, 2.01it/s]
Page 5 in progress...: 81%|██████████|
| 29/36 [00:12<00:03, 2.20it/s]
Page 5 in progress...: 83%|██████████|
| 30/36 [00:12<00:02, 2.27it/s]
Page 5 in progress...: 86%|██████████|
| 31/36 [00:13<00:02, 2.46it/s]
Page 5 in progress...: 89%|██████████|
| 32/36 [00:13<00:01, 2.54it/s]
Page 5 in progress...: 92%|██████████|
| 33/36 [00:13<00:01, 2.57it/s]
Page 5 in progress...: 94%|██████████|
| 34/36 [00:14<00:00, 2.52it/s]
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Page 6 in progress...: 64%|███████████████████████████████████  
| 23/36 [00:09<00:04, 2.78it/s]  
  
Page 6 in progress...: 67%|███████████████████████████████████  
| 24/36 [00:09<00:04, 2.90it/s]  
  
Page 6 in progress...: 69%|███████████████████████████████████  
| 25/36 [00:09<00:04, 2.59it/s]  
  
Page 6 in progress...: 72%|███████████████████████████████████  
| 26/36 [00:10<00:03, 2.79it/s]  
  
Page 6 in progress...: 75%|███████████████████████████████████  
| 27/36 [00:10<00:03, 2.62it/s]  
  
Page 6 in progress...: 78%|███████████████████████████████████  
| 28/36 [00:11<00:03, 2.55it/s]  
  
Page 6 in progress...: 81%|███████████████████████████████████  
| 29/36 [00:11<00:02, 2.51it/s]  
  
Page 6 in progress...: 83%|███████████████████████████████████  
| 30/36 [00:11<00:02, 2.67it/s]  
  
Page 6 in progress...: 86%|███████████████████████████████████  
| 31/36 [00:12<00:01, 2.65it/s]  
  
Page 6 in progress...: 89%|███████████████████████████████████  
| 32/36 [00:12<00:01, 2.58it/s]  
  
Page 6 in progress...: 92%|███████████████████████████████████  
| 33/36 [00:12<00:01, 2.66it/s]  
  
Page 6 in progress...: 94%|███████████████████████████████████  
| 34/36 [00:13<00:00, 2.67it/s]  
  
Page 6 in progress...: 97%|███████████████████████████████████  
| 35/36 [00:13<00:00, 2.72it/s]  
  
Page 6 in progress...: 100%|███████████████████████████████████  
| 36/36 [00:14<00:00, 2.67it/s]  
  
Page 6 in progress...: 100%|███████████████████████████████████  
| 36/36 [00:14<00:00, 2.57it/s]  
  
Page 7 in progress...: 0%|  
| 0/36 [00:00<?, ?it/s]  
  
Page 7 in progress...: 3%|████  
| 1/36 [00:00<00:21, 1.65it/s]  
  
Page 7 in progress...: 6%|████  
| 2/36 [00:01<00:17, 1.98it/s]  
  
Page 7 in progress...: 8%|████  
| 3/36 [00:01<00:14, 2.20it/s]  
  
Page 7 in progress...: 11%|██████  
| 4/36 [00:01<00:14, 2.28it/s]  
  
Page 7 in progress...: 14%|███████  
| 5/36 [00:02<00:13, 2.38it/s]  
  
Page 7 in progress...: 17%|████████  
| 6/36 [00:02<00:11, 2.58it/s]  
  
Page 7 in progress...: 19%|█████████  
| 7/36 [00:02<00:11, 2.57it/s]  
  
Page 7 in progress...: 22%|██████████  
| 8/36 [00:03<00:11, 2.54it/s]  
  
Page 7 in progress...: 25%|███████████  
| 9/36 [00:03<00:10, 2.64it/s]  
  
Page 7 in progress...: 28%|███████████  
| 10/36 [00:04<00:09, 2.80it/s]
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Page 7 in progress...: 31%|██████████|
| 11/36 [00:04<00:08, 2.93it/s]
Page 7 in progress...: 33%|███████████|
| 12/36 [00:04<00:07, 3.06it/s]
Page 7 in progress...: 36%|███████████|
| 13/36 [00:05<00:08, 2.78it/s]
Page 7 in progress...: 39%|███████████|
| 14/36 [00:05<00:08, 2.53it/s]
Page 7 in progress...: 42%|███████████|
| 15/36 [00:05<00:08, 2.52it/s]
Page 7 in progress...: 44%|███████████|
| 16/36 [00:06<00:08, 2.31it/s]
Page 7 in progress...: 47%|███████████|
| 17/36 [00:06<00:08, 2.32it/s]
Page 7 in progress...: 50%|███████████|
| 18/36 [00:07<00:07, 2.41it/s]
Page 7 in progress...: 53%|███████████|
| 19/36 [00:07<00:07, 2.34it/s]
Page 7 in progress...: 56%|███████████|
| 20/36 [00:08<00:06, 2.41it/s]
Page 7 in progress...: 58%|███████████|
| 21/36 [00:08<00:05, 2.59it/s]
Page 7 in progress...: 61%|███████████|
| 22/36 [00:08<00:05, 2.70it/s]
Page 7 in progress...: 64%|███████████|
| 23/36 [00:09<00:04, 2.70it/s]
Page 7 in progress...: 67%|███████████|
| 24/36 [00:09<00:04, 2.48it/s]
Page 7 in progress...: 69%|███████████|
| 25/36 [00:09<00:04, 2.49it/s]
Page 7 in progress...: 72%|███████████|
| 26/36 [00:10<00:03, 2.52it/s]
Page 7 in progress...: 75%|███████████|
| 27/36 [00:10<00:03, 2.45it/s]
Page 7 in progress...: 78%|███████████|
| 28/36 [00:11<00:03, 2.36it/s]
Page 7 in progress...: 81%|███████████|
| 29/36 [00:11<00:03, 2.31it/s]
Page 7 in progress...: 83%|███████████|
| 30/36 [00:12<00:02, 2.33it/s]
Page 7 in progress...: 86%|███████████|
| 31/36 [00:12<00:02, 2.46it/s]
Page 7 in progress...: 89%|███████████|
| 32/36 [00:12<00:01, 2.34it/s]
Page 7 in progress...: 92%|███████████|
| 33/36 [00:13<00:01, 2.51it/s]
Page 7 in progress...: 94%|███████████|
| 34/36 [00:13<00:00, 2.57it/s]
Page 7 in progress...: 97%|███████████|
| 35/36 [00:14<00:00, 2.65it/s]
Page 7 in progress...: 100%|███████████|
| 36/36 [00:14<00:00, 2.61it/s]
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Page 8 in progress...: 69%|██████████████████████████████████████████|
| 25/36 [00:10<00:04, 2.21it/s]
Page 8 in progress...: 72%|██████████████████████████████████████████|
| 26/36 [00:10<00:04, 2.39it/s]
Page 8 in progress...: 75%|██████████████████████████████████████████|
| 27/36 [00:11<00:03, 2.30it/s]
Page 8 in progress...: 78%|██████████████████████████████████████████|
| 28/36 [00:11<00:03, 2.29it/s]
Page 8 in progress...: 81%|██████████████████████████████████████████|
| 29/36 [00:12<00:03, 2.32it/s]
Page 8 in progress...: 83%|██████████████████████████████████████████|
| 30/36 [00:12<00:02, 2.18it/s]
Page 8 in progress...: 86%|██████████████████████████████████████████|
| 31/36 [00:13<00:02, 2.16it/s]
Page 8 in progress...: 89%|██████████████████████████████████████████|
| 32/36 [00:13<00:01, 2.26it/s]
Page 8 in progress...: 92%|██████████████████████████████████████████|
| 33/36 [00:13<00:01, 2.33it/s]
Page 8 in progress...: 94%|██████████████████████████████████████████|
| 34/36 [00:14<00:00, 2.30it/s]
Page 8 in progress...: 97%|██████████████████████████████████████████|
| 35/36 [00:14<00:00, 2.24it/s]
Page 8 in progress...: 100%|██████████████████████████████████████████|
| 36/36 [00:15<00:00, 2.11it/s]
Page 8 in progress...: 100%|██████████████████████████████████████████|
| 36/36 [00:15<00:00, 2.33it/s]
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| 0/36 [00:00<?, ?it/s]
Page 9 in progress...: 3%|██████|
| 1/36 [00:00<00:34, 1.01it/s]
Page 9 in progress...: 6%|██████|
| 2/36 [00:01<00:21, 1.61it/s]
Page 9 in progress...: 8%|██████|
| 3/36 [00:01<00:17, 1.94it/s]
Page 9 in progress...: 11%|██████|
| 4/36 [00:02<00:14, 2.22it/s]
Page 9 in progress...: 14%|██████|
| 5/36 [00:02<00:13, 2.38it/s]
Page 9 in progress...: 17%|██████|
| 6/36 [00:02<00:12, 2.31it/s]
Page 9 in progress...: 19%|██████|
| 7/36 [00:03<00:11, 2.44it/s]
Page 9 in progress...: 22%|██████|
| 8/36 [00:03<00:12, 2.22it/s]
Page 9 in progress...: 25%|██████|
| 9/36 [00:04<00:12, 2.24it/s]
Page 9 in progress...: 28%|██████|
| 10/36 [00:04<00:11, 2.18it/s]
Page 9 in progress...: 31%|██████|
| 11/36 [00:05<00:11, 2.26it/s]
Page 9 in progress...: 33%|██████|
| 12/36 [00:05<00:11, 2.18it/s]
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Page 9 in progress...: 36%|███████████  
| 13/36 [00:06<00:10, 2.24it/s]  
Page 9 in progress...: 39%|███████████  
| 14/36 [00:06<00:10, 2.12it/s]  
Page 9 in progress...: 42%|███████████  
| 15/36 [00:07<00:09, 2.20it/s]  
Page 9 in progress...: 44%|███████████  
| 16/36 [00:07<00:09, 2.13it/s]  
Page 9 in progress...: 47%|███████████  
| 17/36 [00:08<00:09, 1.96it/s]  
Page 9 in progress...: 50%|███████████  
| 18/36 [00:08<00:08, 2.02it/s]  
Page 9 in progress...: 53%|███████████  
| 19/36 [00:08<00:07, 2.17it/s]  
Page 9 in progress...: 56%|███████████  
| 20/36 [00:09<00:07, 2.17it/s]  
Page 9 in progress...: 58%|███████████  
| 21/36 [00:09<00:06, 2.15it/s]  
Page 9 in progress...: 61%|███████████  
| 22/36 [00:10<00:06, 2.19it/s]  
Page 9 in progress...: 64%|███████████  
| 23/36 [00:10<00:05, 2.20it/s]  
Page 9 in progress...: 67%|███████████  
| 24/36 [00:11<00:05, 2.29it/s]  
Page 9 in progress...: 69%|███████████  
| 25/36 [00:11<00:04, 2.40it/s]  
Page 9 in progress...: 72%|███████████  
| 26/36 [00:12<00:04, 2.29it/s]  
Page 9 in progress...: 75%|███████████  
| 27/36 [00:12<00:03, 2.30it/s]  
Page 9 in progress...: 78%|███████████  
| 28/36 [00:12<00:03, 2.34it/s]  
Page 9 in progress...: 81%|███████████  
| 29/36 [00:13<00:03, 2.33it/s]  
Page 9 in progress...: 83%|███████████  
| 30/36 [00:13<00:02, 2.20it/s]  
Page 9 in progress...: 86%|███████████  
| 31/36 [00:14<00:02, 2.17it/s]  
Page 9 in progress...: 89%|███████████  
| 32/36 [00:14<00:01, 2.17it/s]  
Page 9 in progress...: 92%|███████████  
| 33/36 [00:15<00:02, 1.49it/s]  
Page 9 in progress...: 94%|███████████  
| 34/36 [00:16<00:01, 1.71it/s]  
Page 9 in progress...: 97%|███████████  
| 35/36 [00:16<00:00, 1.93it/s]  
Page 9 in progress...: 100%|███████████  
| 36/36 [00:17<00:00, 2.03it/s]  
Page 9 in progress...: 100%|███████████  
| 36/36 [00:17<00:00, 2.11it/s]  
  
Page 10 in progress...: 0%|  
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Page 10 in progress...: 3%|███
| 1/36 [00:00<00:25, 1.35it/s]
Page 10 in progress...: 6%|████
| 2/36 [00:01<00:20, 1.64it/s]
Page 10 in progress...: 8%|██████
| 3/36 [00:01<00:17, 1.88it/s]
Page 10 in progress...: 11%|███████
| 4/36 [00:02<00:14, 2.15it/s]
Page 10 in progress...: 14%|██████████
| 5/36 [00:02<00:13, 2.29it/s]
Page 10 in progress...: 17%|███████████
| 6/36 [00:02<00:11, 2.51it/s]
Page 10 in progress...: 19%|████████████
| 7/36 [00:03<00:11, 2.45it/s]
Page 10 in progress...: 22%|█████████████
| 8/36 [00:03<00:10, 2.56it/s]
Page 10 in progress...: 25%|██████████████
| 9/36 [00:03<00:10, 2.50it/s]
Page 10 in progress...: 28%|███████████████
| 10/36 [00:04<00:10, 2.49it/s]
Page 10 in progress...: 31%|████████████████
| 11/36 [00:04<00:10, 2.43it/s]
Page 10 in progress...: 33%|█████████████████
| 12/36 [00:05<00:09, 2.66it/s]
Page 10 in progress...: 36%|██████████████████
| 13/36 [00:05<00:08, 2.71it/s]
Page 10 in progress...: 39%|███████████████████
| 14/36 [00:05<00:07, 2.80it/s]
Page 10 in progress...: 42%|████████████████████
| 15/36 [00:06<00:07, 2.76it/s]
Page 10 in progress...: 44%|█████████████████████
| 16/36 [00:06<00:07, 2.74it/s]
Page 10 in progress...: 47%|██████████████████████
| 17/36 [00:06<00:06, 2.74it/s]
Page 10 in progress...: 50%|███████████████████████
| 18/36 [00:07<00:06, 2.75it/s]
Page 10 in progress...: 53%|████████████████████████
| 19/36 [00:07<00:06, 2.61it/s]
Page 10 in progress...: 56%|█████████████████████████
| 20/36 [00:08<00:06, 2.30it/s]
Page 10 in progress...: 58%|██████████████████████████
| 21/36 [00:08<00:06, 2.39it/s]
Page 10 in progress...: 61%|███████████████████████████
| 22/36 [00:09<00:06, 2.15it/s]
Page 10 in progress...: 64%|████████████████████████████
| 23/36 [00:09<00:05, 2.41it/s]
Page 10 in progress...: 67%|█████████████████████████████
| 24/36 [00:09<00:05, 2.39it/s]
Page 10 in progress...: 69%|██████████████████████████████
| 25/36 [00:10<00:04, 2.40it/s]
Page 10 in progress...: 72%|███████████████████████████████
| 26/36 [00:10<00:03, 2.54it/s]
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Page 10 in progress...: 75%|███████████  
| 27/36 [00:11<00:03, 2.55it/s]  
Page 10 in progress...: 78%|███████████  
| 28/36 [00:11<00:03, 2.54it/s]  
Page 10 in progress...: 81%|███████████  
| 29/36 [00:11<00:02, 2.73it/s]  
Page 10 in progress...: 83%|███████████  
| 30/36 [00:12<00:02, 2.63it/s]  
Page 10 in progress...: 86%|███████████  
| 31/36 [00:12<00:01, 2.51it/s]  
Page 10 in progress...: 89%|███████████  
| 32/36 [00:13<00:01, 2.44it/s]  
Page 10 in progress...: 92%|███████████  
| 33/36 [00:13<00:01, 2.51it/s]  
Page 10 in progress...: 94%|███████████  
| 34/36 [00:14<00:01, 1.65it/s]  
Page 10 in progress...: 97%|███████████  
| 35/36 [00:14<00:00, 1.85it/s]  
Page 10 in progress...: 100%|███████████  
| 36/36 [00:15<00:00, 2.13it/s]  
Page 10 in progress...: 100%|███████████  
| 36/36 [00:15<00:00, 2.37it/s]  
  
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| 0/36 [00:00<?, ?it/s]  
Page 11 in progress...: 3%|███  
| 1/36 [00:00<00:22, 1.53it/s]  
Page 11 in progress...: 6%|█████  
| 2/36 [00:01<00:20, 1.63it/s]  
Page 11 in progress...: 8%|█████  
| 3/36 [00:01<00:16, 1.96it/s]  
Page 11 in progress...: 11%|███████  
| 4/36 [00:01<00:14, 2.23it/s]  
Page 11 in progress...: 14%|█████████  
| 5/36 [00:02<00:13, 2.34it/s]  
Page 11 in progress...: 17%|█████████  
| 6/36 [00:02<00:12, 2.48it/s]  
Page 11 in progress...: 19%|█████████  
| 7/36 [00:03<00:11, 2.44it/s]  
Page 11 in progress...: 22%|█████████  
| 8/36 [00:03<00:11, 2.47it/s]  
Page 11 in progress...: 25%|█████████  
| 9/36 [00:03<00:10, 2.56it/s]  
Page 11 in progress...: 28%|█████████  
| 10/36 [00:04<00:09, 2.69it/s]  
Page 11 in progress...: 31%|█████████  
| 11/36 [00:04<00:09, 2.57it/s]  
Page 11 in progress...: 33%|█████████  
| 12/36 [00:05<00:09, 2.50it/s]  
Page 11 in progress...: 36%|█████████  
| 13/36 [00:05<00:09, 2.47it/s]  
Page 11 in progress...: 39%|█████████  
| 14/36 [00:05<00:08, 2.54it/s]
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Page 11 in progress...: 42%|███████████████████████  
| 15/36 [00:06<00:08, 2.59it/s]  
Page 11 in progress...: 44%|███████████████████████  
| 16/36 [00:06<00:07, 2.52it/s]  
Page 11 in progress...: 47%|███████████████████████  
| 17/36 [00:07<00:07, 2.51it/s]  
Page 11 in progress...: 50%|███████████████████████  
| 18/36 [00:07<00:06, 2.59it/s]  
Page 11 in progress...: 53%|███████████████████████  
| 19/36 [00:07<00:06, 2.52it/s]  
Page 11 in progress...: 56%|███████████████████████  
| 20/36 [00:08<00:06, 2.46it/s]  
Page 11 in progress...: 58%|███████████████████████  
| 21/36 [00:08<00:06, 2.45it/s]  
Page 11 in progress...: 61%|███████████████████████  
| 22/36 [00:09<00:05, 2.56it/s]  
Page 11 in progress...: 64%|███████████████████████  
| 23/36 [00:09<00:04, 2.78it/s]  
Page 11 in progress...: 67%|███████████████████████  
| 24/36 [00:09<00:04, 2.86it/s]  
Page 11 in progress...: 69%|███████████████████████  
| 25/36 [00:09<00:03, 2.94it/s]  
Page 11 in progress...: 72%|███████████████████████  
| 26/36 [00:10<00:03, 2.71it/s]  
Page 11 in progress...: 75%|███████████████████████  
| 27/36 [00:10<00:03, 2.62it/s]  
Page 11 in progress...: 78%|███████████████████████  
| 28/36 [00:11<00:02, 2.74it/s]  
Page 11 in progress...: 81%|███████████████████████  
| 29/36 [00:11<00:02, 2.71it/s]  
Page 11 in progress...: 83%|███████████████████████  
| 30/36 [00:11<00:02, 2.63it/s]  
Page 11 in progress...: 86%|███████████████████████  
| 31/36 [00:12<00:01, 2.82it/s]  
Page 11 in progress...: 89%|███████████████████████  
| 32/36 [00:12<00:01, 2.72it/s]  
Page 11 in progress...: 92%|███████████████████████  
| 33/36 [00:12<00:01, 2.84it/s]  
Page 11 in progress...: 94%|███████████████████████  
| 34/36 [00:13<00:00, 2.60it/s]  
Page 11 in progress...: 97%|███████████████████████  
| 35/36 [00:13<00:00, 2.84it/s]  
Page 11 in progress...: 100%|███████████████████████  
| 36/36 [00:14<00:00, 2.78it/s]  
Page 11 in progress...: 100%|███████████████████████  
| 36/36 [00:14<00:00, 2.56it/s]  
  
Page 12 in progress...: 0%|  
| 0/36 [00:00<?, ?it/s]  
  
Page 12 in progress...: 3%|███  
| 1/36 [00:00<00:14, 2.49it/s]  
  
Page 12 in progress...: 6%|█████  
| 2/36 [00:00<00:17, 1.95it/s]
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Page 12 in progress...: 8%|██████|
| 3/36 [00:02<00:25, 1.29it/s]
Page 12 in progress...: 11%|███████|
| 4/36 [00:02<00:18, 1.71it/s]
Page 12 in progress...: 14%|████████|
| 5/36 [00:02<00:16, 1.90it/s]
Page 12 in progress...: 17%|█████████|
| 6/36 [00:03<00:13, 2.20it/s]
Page 12 in progress...: 19%|██████████|
| 7/36 [00:03<00:11, 2.44it/s]
Page 12 in progress...: 22%|███████████|
| 8/36 [00:03<00:11, 2.52it/s]
Page 12 in progress...: 25%|████████████|
| 9/36 [00:04<00:10, 2.49it/s]
Page 12 in progress...: 28%|█████████████|
| 10/36 [00:04<00:10, 2.54it/s]
Page 12 in progress...: 31%|██████████████|
| 11/36 [00:05<00:10, 2.45it/s]
Page 12 in progress...: 33%|███████████████|
| 12/36 [00:05<00:10, 2.18it/s]
Page 12 in progress...: 36%|████████████████|
| 13/36 [00:06<00:10, 2.19it/s]
Page 12 in progress...: 39%|█████████████████|
| 14/36 [00:06<00:09, 2.36it/s]
Page 12 in progress...: 42%|██████████████████|
| 15/36 [00:06<00:08, 2.42it/s]
Page 12 in progress...: 44%|███████████████████|
| 16/36 [00:07<00:07, 2.53it/s]
Page 12 in progress...: 47%|████████████████████|
| 17/36 [00:07<00:07, 2.48it/s]
Page 12 in progress...: 50%|█████████████████████|
| 18/36 [00:07<00:06, 2.59it/s]
Page 12 in progress...: 53%|██████████████████████|
| 19/36 [00:08<00:06, 2.66it/s]
Page 12 in progress...: 56%|███████████████████████|
| 20/36 [00:08<00:06, 2.56it/s]
Page 12 in progress...: 58%|████████████████████████|
| 21/36 [00:09<00:06, 2.49it/s]
Page 12 in progress...: 61%|█████████████████████████|
| 22/36 [00:09<00:06, 2.30it/s]
Page 12 in progress...: 64%|██████████████████████████|
| 23/36 [00:09<00:05, 2.42it/s]
Page 12 in progress...: 67%|███████████████████████████|
| 24/36 [00:10<00:05, 2.22it/s]
Page 12 in progress...: 69%|████████████████████████████|
| 25/36 [00:10<00:04, 2.28it/s]
Page 12 in progress...: 72%|█████████████████████████████|
| 26/36 [00:11<00:04, 2.37it/s]
Page 12 in progress...: 75%|██████████████████████████████|
| 27/36 [00:11<00:03, 2.51it/s]
Page 12 in progress...: 78%|███████████████████████████████|
| 28/36 [00:12<00:03, 2.62it/s]
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Page 12 in progress...: 81%|███████████  
| 29/36 [00:12<00:02, 2.55it/s]  
  
Page 12 in progress...: 83%|███████████  
| 30/36 [00:12<00:02, 2.64it/s]  
  
Page 12 in progress...: 86%|███████████  
| 31/36 [00:13<00:01, 2.68it/s]  
  
Page 12 in progress...: 89%|███████████  
| 32/36 [00:13<00:01, 2.65it/s]  
  
Page 12 in progress...: 92%|███████████  
| 33/36 [00:13<00:01, 2.72it/s]  
  
Page 12 in progress...: 94%|███████████  
| 34/36 [00:14<00:00, 2.74it/s]  
  
Page 12 in progress...: 97%|███████████  
| 35/36 [00:14<00:00, 2.64it/s]  
  
Page 12 in progress...: 100%|███████████  
| 36/36 [00:15<00:00, 2.62it/s]  
  
Page 12 in progress...: 100%|███████████  
| 36/36 [00:15<00:00, 2.40it/s]  
  
Page 13 in progress...: 0%|  
| 0/36 [00:00<?, ?it/s]  
  
Page 13 in progress...: 3%|█████  
| 1/36 [00:00<00:21, 1.61it/s]  
  
Page 13 in progress...: 6%|█████  
| 2/36 [00:01<00:18, 1.81it/s]  
  
Page 13 in progress...: 8%|█████  
| 3/36 [00:01<00:14, 2.23it/s]  
  
Page 13 in progress...: 11%|█████  
| 4/36 [00:02<00:23, 1.35it/s]  
  
Page 13 in progress...: 14%|█████  
| 5/36 [00:02<00:18, 1.67it/s]  
  
Page 13 in progress...: 17%|█████  
| 6/36 [00:03<00:16, 1.84it/s]  
  
Page 13 in progress...: 19%|█████  
| 7/36 [00:03<00:14, 2.06it/s]  
  
Page 13 in progress...: 22%|█████  
| 8/36 [00:04<00:13, 2.08it/s]  
  
Page 13 in progress...: 25%|█████  
| 9/36 [00:04<00:11, 2.29it/s]  
  
Page 13 in progress...: 28%|█████  
| 10/36 [00:04<00:10, 2.42it/s]  
  
Page 13 in progress...: 31%|█████  
| 11/36 [00:05<00:09, 2.56it/s]  
  
Page 13 in progress...: 33%|█████  
| 12/36 [00:05<00:09, 2.66it/s]  
  
Page 13 in progress...: 36%|█████  
| 13/36 [00:06<00:08, 2.56it/s]  
  
Page 13 in progress...: 39%|█████  
| 14/36 [00:06<00:08, 2.45it/s]  
  
Page 13 in progress...: 42%|█████  
| 15/36 [00:06<00:08, 2.52it/s]  
  
Page 13 in progress...: 44%|█████  
| 16/36 [00:07<00:07, 2.63it/s]
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Page 13 in progress...: 47%|███████████  
| 17/36 [00:07<00:07, 2.47it/s]  
  
Page 13 in progress...: 50%|███████████  
| 18/36 [00:08<00:07, 2.51it/s]  
  
Page 13 in progress...: 53%|███████████  
| 19/36 [00:08<00:06, 2.46it/s]  
  
Page 13 in progress...: 56%|███████████  
| 20/36 [00:08<00:06, 2.43it/s]  
  
Page 13 in progress...: 58%|███████████  
| 21/36 [00:09<00:06, 2.45it/s]  
  
Page 13 in progress...: 61%|███████████  
| 22/36 [00:09<00:05, 2.65it/s]  
  
Page 13 in progress...: 64%|███████████  
| 23/36 [00:09<00:04, 2.69it/s]  
  
Page 13 in progress...: 67%|███████████  
| 24/36 [00:10<00:04, 2.70it/s]  
  
Page 13 in progress...: 69%|███████████  
| 25/36 [00:10<00:03, 2.82it/s]  
  
Page 13 in progress...: 72%|███████████  
| 26/36 [00:10<00:03, 2.92it/s]  
  
Page 13 in progress...: 75%|███████████  
| 27/36 [00:11<00:03, 2.93it/s]  
  
Page 13 in progress...: 78%|███████████  
| 28/36 [00:11<00:02, 2.90it/s]  
  
Page 13 in progress...: 81%|███████████  
| 29/36 [00:12<00:02, 2.93it/s]  
  
Page 13 in progress...: 83%|███████████  
| 30/36 [00:12<00:02, 2.89it/s]  
  
Page 13 in progress...: 86%|███████████  
| 31/36 [00:12<00:01, 2.84it/s]  
  
Page 13 in progress...: 89%|███████████  
| 32/36 [00:13<00:01, 2.63it/s]  
  
Page 13 in progress...: 92%|███████████  
| 33/36 [00:13<00:01, 2.55it/s]  
  
Page 13 in progress...: 94%|███████████  
| 34/36 [00:13<00:00, 2.55it/s]  
  
Page 13 in progress...: 97%|███████████  
| 35/36 [00:14<00:00, 2.55it/s]  
  
Page 13 in progress...: 100%|███████████  
| 36/36 [00:14<00:00, 2.64it/s]  
  
Page 13 in progress...: 100%|███████████  
| 36/36 [00:14<00:00, 2.44it/s]  
  
Page 14 in progress...: 0%|  
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Page 14 in progress...: 3%|███  
| 1/36 [00:00<00:30, 1.13it/s]  
  
Page 14 in progress...: 6%|█████  
| 2/36 [00:01<00:20, 1.69it/s]  
  
Page 14 in progress...: 8%|███████  
| 3/36 [00:01<00:16, 2.04it/s]  
  
Page 14 in progress...: 11%|███████  
| 4/36 [00:02<00:14, 2.17it/s]
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Page 14 in progress...: 14%|██████████|
| 5/36 [00:02<00:13, 2.33it/s]
Page 14 in progress...: 17%|███████████|
| 6/36 [00:02<00:12, 2.36it/s]
Page 14 in progress...: 19%|███████████|
| 7/36 [00:03<00:11, 2.50it/s]
Page 14 in progress...: 22%|███████████|
| 8/36 [00:03<00:10, 2.56it/s]
Page 14 in progress...: 25%|███████████|
| 9/36 [00:03<00:10, 2.50it/s]
Page 14 in progress...: 28%|███████████|
| 10/36 [00:04<00:09, 2.65it/s]
Page 14 in progress...: 31%|███████████|
| 11/36 [00:04<00:09, 2.71it/s]
Page 14 in progress...: 33%|███████████|
| 12/36 [00:05<00:08, 2.74it/s]
Page 14 in progress...: 36%|███████████|
| 13/36 [00:05<00:08, 2.60it/s]
Page 14 in progress...: 39%|███████████|
| 14/36 [00:05<00:08, 2.46it/s]
Page 14 in progress...: 42%|███████████|
| 15/36 [00:06<00:08, 2.42it/s]
Page 14 in progress...: 44%|███████████|
| 16/36 [00:06<00:08, 2.40it/s]
Page 14 in progress...: 47%|███████████|
| 17/36 [00:07<00:07, 2.53it/s]
Page 14 in progress...: 50%|███████████|
| 18/36 [00:07<00:07, 2.54it/s]
Page 14 in progress...: 53%|███████████|
| 19/36 [00:07<00:06, 2.64it/s]
Page 14 in progress...: 56%|███████████|
| 20/36 [00:08<00:05, 2.69it/s]
Page 14 in progress...: 58%|███████████|
| 21/36 [00:08<00:06, 2.39it/s]
Page 14 in progress...: 61%|███████████|
| 22/36 [00:09<00:05, 2.51it/s]
Page 14 in progress...: 64%|███████████|
| 23/36 [00:09<00:04, 2.64it/s]
Page 14 in progress...: 67%|███████████|
| 24/36 [00:09<00:04, 2.73it/s]
Page 14 in progress...: 69%|███████████|
| 25/36 [00:10<00:04, 2.65it/s]
Page 14 in progress...: 72%|███████████|
| 26/36 [00:10<00:03, 2.69it/s]
Page 14 in progress...: 75%|███████████|
| 27/36 [00:10<00:03, 2.57it/s]
Page 14 in progress...: 78%|███████████|
| 28/36 [00:11<00:03, 2.59it/s]
Page 14 in progress...: 81%|███████████|
| 29/36 [00:11<00:02, 2.61it/s]
Page 14 in progress...: 83%|███████████|
| 30/36 [00:11<00:02, 2.80it/s]
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Page 14 in progress...: 86%|███████████  
| 31/36 [00:12<00:01, 2.66it/s]  
  
Page 14 in progress...: 89%|███████████  
| 32/36 [00:12<00:01, 2.48it/s]  
  
Page 14 in progress...: 92%|███████████  
| 33/36 [00:13<00:01, 2.46it/s]  
  
Page 14 in progress...: 94%|███████████  
| 34/36 [00:13<00:00, 2.61it/s]  
  
Page 14 in progress...: 97%|███████████  
| 35/36 [00:14<00:00, 2.56it/s]  
  
Page 14 in progress...: 100%|███████████  
| 36/36 [00:14<00:00, 2.60it/s]  
  
Page 14 in progress...: 100%|███████████  
| 36/36 [00:14<00:00, 2.50it/s]  
  
Page 15 in progress...: 0%|  
| 0/36 [00:00<?, ?it/s]  
  
Page 15 in progress...: 3%|███  
| 1/36 [00:00<00:19, 1.77it/s]  
  
Page 15 in progress...: 6%|█████  
| 2/36 [00:01<00:19, 1.73it/s]  
  
Page 15 in progress...: 8%|███████  
| 3/36 [00:01<00:16, 1.96it/s]  
  
Page 15 in progress...: 11%|█████████  
| 4/36 [00:02<00:16, 1.93it/s]  
  
Page 15 in progress...: 14%|███████████  
| 5/36 [00:02<00:14, 2.14it/s]  
  
Page 15 in progress...: 17%|███████████  
| 6/36 [00:02<00:12, 2.38it/s]  
  
Page 15 in progress...: 19%|███████████  
| 7/36 [00:03<00:11, 2.42it/s]  
  
Page 15 in progress...: 22%|███████████  
| 8/36 [00:03<00:11, 2.50it/s]  
  
Page 15 in progress...: 25%|███████████  
| 9/36 [00:03<00:10, 2.49it/s]  
  
Page 15 in progress...: 28%|███████████  
| 10/36 [00:04<00:10, 2.44it/s]  
  
Page 15 in progress...: 31%|███████████  
| 11/36 [00:04<00:10, 2.42it/s]  
  
Page 15 in progress...: 33%|███████████  
| 12/36 [00:05<00:10, 2.29it/s]  
  
Page 15 in progress...: 36%|███████████  
| 13/36 [00:05<00:10, 2.26it/s]  
  
Page 15 in progress...: 39%|███████████  
| 14/36 [00:06<00:09, 2.33it/s]  
  
Page 15 in progress...: 42%|███████████  
| 15/36 [00:06<00:09, 2.29it/s]  
  
Page 15 in progress...: 44%|███████████  
| 16/36 [00:07<00:08, 2.38it/s]  
  
Page 15 in progress...: 47%|███████████  
| 17/36 [00:07<00:08, 2.30it/s]  
  
Page 15 in progress...: 50%|███████████  
| 18/36 [00:07<00:07, 2.27it/s]
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Page 15 in progress...: 53%|███████████████████████████████  
| 19/36 [00:08<00:07, 2.39it/s]  
  
Page 15 in progress...: 56%|███████████████████████████████  
| 20/36 [00:08<00:06, 2.58it/s]  
  
Page 15 in progress...: 58%|███████████████████████████████  
| 21/36 [00:09<00:05, 2.57it/s]  
  
Page 15 in progress...: 61%|███████████████████████████████  
| 22/36 [00:09<00:06, 2.30it/s]  
  
Page 15 in progress...: 64%|███████████████████████████████  
| 23/36 [00:09<00:05, 2.34it/s]  
  
Page 15 in progress...: 67%|███████████████████████████████  
| 24/36 [00:10<00:04, 2.56it/s]  
  
Page 15 in progress...: 69%|███████████████████████████████  
| 25/36 [00:10<00:04, 2.69it/s]  
  
Page 15 in progress...: 72%|███████████████████████████████  
| 26/36 [00:11<00:03, 2.52it/s]  
  
Page 15 in progress...: 75%|███████████████████████████████  
| 27/36 [00:11<00:03, 2.68it/s]  
  
Page 15 in progress...: 78%|███████████████████████████████  
| 28/36 [00:11<00:02, 2.73it/s]  
  
Page 15 in progress...: 81%|███████████████████████████████  
| 29/36 [00:12<00:02, 2.79it/s]  
  
Page 15 in progress...: 83%|███████████████████████████████  
| 30/36 [00:12<00:02, 2.69it/s]  
  
Page 15 in progress...: 86%|███████████████████████████████  
| 31/36 [00:12<00:01, 2.67it/s]  
  
Page 15 in progress...: 89%|███████████████████████████████  
| 32/36 [00:13<00:01, 2.56it/s]  
  
Page 15 in progress...: 92%|███████████████████████████████  
| 33/36 [00:13<00:01, 2.69it/s]  
  
Page 15 in progress...: 94%|███████████████████████████████  
| 34/36 [00:13<00:00, 2.80it/s]  
  
Page 15 in progress...: 97%|███████████████████████████████  
| 35/36 [00:14<00:00, 2.90it/s]  
  
Page 15 in progress...: 100%|███████████████████████████████  
| 36/36 [00:14<00:00, 2.73it/s]  
  
Page 15 in progress...: 100%|███████████████████████████████  
| 36/36 [00:14<00:00, 2.46it/s]  
  
Page 16 in progress...: 0%|  
| 0/36 [00:00<?, ?it/s]  
  
Page 16 in progress...: 3%|███  
| 1/36 [00:00<00:18, 1.86it/s]  
  
Page 16 in progress...: 6%|█████  
| 2/36 [00:01<00:17, 1.97it/s]  
  
Page 16 in progress...: 8%|███████  
| 3/36 [00:01<00:14, 2.29it/s]  
  
Page 16 in progress...: 11%|█████████  
| 4/36 [00:01<00:13, 2.36it/s]  
  
Page 16 in progress...: 14%|███████████  
| 5/36 [00:02<00:12, 2.57it/s]  
  
Page 16 in progress...: 17%|█████████████  
| 6/36 [00:02<00:11, 2.66it/s]
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Page 16 in progress...: 19%|██████████|
| 7/36 [00:02<00:10, 2.71it/s]
Page 16 in progress...: 22%|██████████|
| 8/36 [00:03<00:10, 2.67it/s]
Page 16 in progress...: 25%|██████████|
| 9/36 [00:03<00:10, 2.69it/s]
Page 16 in progress...: 28%|██████████|
| 10/36 [00:03<00:09, 2.80it/s]
Page 16 in progress...: 31%|██████████|
| 11/36 [00:04<00:09, 2.76it/s]
Page 16 in progress...: 33%|██████████|
| 12/36 [00:04<00:09, 2.58it/s]
Page 16 in progress...: 36%|██████████|
| 13/36 [00:05<00:09, 2.53it/s]
Page 16 in progress...: 39%|██████████|
| 14/36 [00:05<00:08, 2.45it/s]
Page 16 in progress...: 42%|██████████|
| 15/36 [00:05<00:08, 2.51it/s]
Page 16 in progress...: 44%|██████████|
| 16/36 [00:06<00:07, 2.53it/s]
Page 16 in progress...: 47%|██████████|
| 17/36 [00:06<00:07, 2.63it/s]
Page 16 in progress...: 50%|██████████|
| 18/36 [00:07<00:07, 2.57it/s]
Page 16 in progress...: 53%|██████████|
| 19/36 [00:07<00:06, 2.54it/s]
Page 16 in progress...: 56%|██████████|
| 20/36 [00:07<00:06, 2.60it/s]
Page 16 in progress...: 58%|██████████|
| 21/36 [00:08<00:05, 2.67it/s]
Page 16 in progress...: 61%|██████████|
| 22/36 [00:08<00:05, 2.69it/s]
Page 16 in progress...: 64%|██████████|
| 23/36 [00:08<00:04, 2.67it/s]
Page 16 in progress...: 67%|██████████|
| 24/36 [00:10<00:07, 1.52it/s]
Page 16 in progress...: 69%|██████████|
| 25/36 [00:10<00:06, 1.82it/s]
Page 16 in progress...: 72%|██████████|
| 26/36 [00:10<00:05, 1.98it/s]
Page 16 in progress...: 75%|██████████|
| 27/36 [00:11<00:04, 2.10it/s]
Page 16 in progress...: 78%|██████████|
| 28/36 [00:11<00:03, 2.07it/s]
Page 16 in progress...: 81%|██████████|
| 29/36 [00:12<00:03, 2.20it/s]
Page 16 in progress...: 83%|██████████|
| 30/36 [00:12<00:02, 2.33it/s]
Page 16 in progress...: 86%|██████████|
| 31/36 [00:13<00:02, 2.36it/s]
Page 16 in progress...: 89%|██████████|
| 32/36 [00:13<00:01, 2.52it/s]
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Page 16 in progress...: 92%|███████████  
| 33/36 [00:13<00:01, 2.56it/s]  
Page 16 in progress...: 94%|███████████  
| 34/36 [00:14<00:00, 2.64it/s]  
Page 16 in progress...: 97%|███████████  
| 35/36 [00:14<00:00, 2.70it/s]  
Page 16 in progress...: 100%|███████████  
| 36/36 [00:14<00:00, 2.72it/s]  
Page 16 in progress...: 100%|███████████  
| 36/36 [00:14<00:00, 2.43it/s]  
  
Page 17 in progress...: 0%|  
| 0/36 [00:00<?, ?it/s]  
Page 17 in progress...: 3%|███  
| 1/36 [00:00<00:16, 2.07it/s]  
Page 17 in progress...: 6%|█████  
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```
In [10]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [11]: df = pd.read_csv("newegg.csv")
          print(df.shape)
          df.head()
```

 $(720, 9)$

Out[11]:

	img	rating	reviews	brand	title
--	-----	--------	---------	-------	-------

0	https://c1.neweggimages.com/productimage/nb300...	4.7	(125)	Sapphire Tech	SAPPHIRE PULSE Radeon RX 90 XT 160 GDDR6 PCIe
1	https://c1.neweggimages.com/productimage/nb300...	4.7	(111)	ASRock	ASRock Challenger Radeon RX 90 XT 160 GDDR6
2	https://c1.neweggimages.com/productimage/nb300...	4.8	(241)	GIGABYTE	GIGABYTE Gaming GeForce RTX 50 Ti 160 GDDR6
3	https://c1.neweggimages.com/productimage/nb300...	4.6	(139)	ASRock	ASRock Challenger Radeon RX 90 XT 160 GDDR6
4	https://c1.neweggimages.com/productimage/nb300...	4.5	(74)	ASRock	ASRock Challenger Radeon RX 90 160 GDDR6 PCIe

```
In [12]: df["brand"] = df["brand"].fillna(df["title"].str.split().str[0])

df["price"] = pd.to_numeric(df["price"].astype(str).str.replace(r"^\0-9.", ""),
df["original_price"] = pd.to_numeric(df["original_price"].astype(str).str.replace(r"^\0-9.", ""))

df["rating"] = pd.to_numeric(df["rating"].astype(str).str.extract(r"([0-9.]+)"))
```

```

df["reviews"] = pd.to_numeric(df["reviews"].astype(str).str.extract(r"(\d+)")[0])

df["memory_gb"] = pd.to_numeric(df["title"].str.extract(r"(\d+)\s*GB")[0], error

t = df["title"].str.upper()
df["series"] = np.select(
    [t.str.contains("ADA"), t.str.contains(r"\bA10\b"),
     t.str.contains(r"RTX\s*50"), t.str.contains(r"RTX\s*40"), t.str.contains(r"
     t.str.contains(r"GTX\s*16"), t.str.contains(r"GTX\s*10"),
     t.str.contains(r"RX\s*9\d{3}"), t.str.contains(r"RX\s*7\d{3}"), t.str.conta
    ["RTX Ada (Pro)", "NVIDIA A-series (Pro)", "RTX 5000", "RTX 4000", "RTX 3000
     "GTX 1600", "GTX 1000", "RX 9000", "RX 7000", "RX 6000"],
    default="Other")

df = df.dropna(subset=["price"]).reset_index(drop=True)
df["price_per_gb"] = (df["price"] / df["memory_gb"]).round(2)

print("Clean rows:", len(df))
df[["brand", "series", "memory_gb", "price", "rating"]].head()

```

Clean rows: 717

Out[12]:

	brand	series	memory_gb	price	rating
0	Sapphire Tech	RX 9000	16.0	499.99	4.7
1	ASRock	RX 9000	16.0	469.99	4.7
2	GIGABYTE	RTX 5000	16.0	1249.99	4.8
3	ASRock	RX 9000	16.0	808.99	4.6
4	ASRock	RX 9000	16.0	649.99	4.5

In [13]:

```

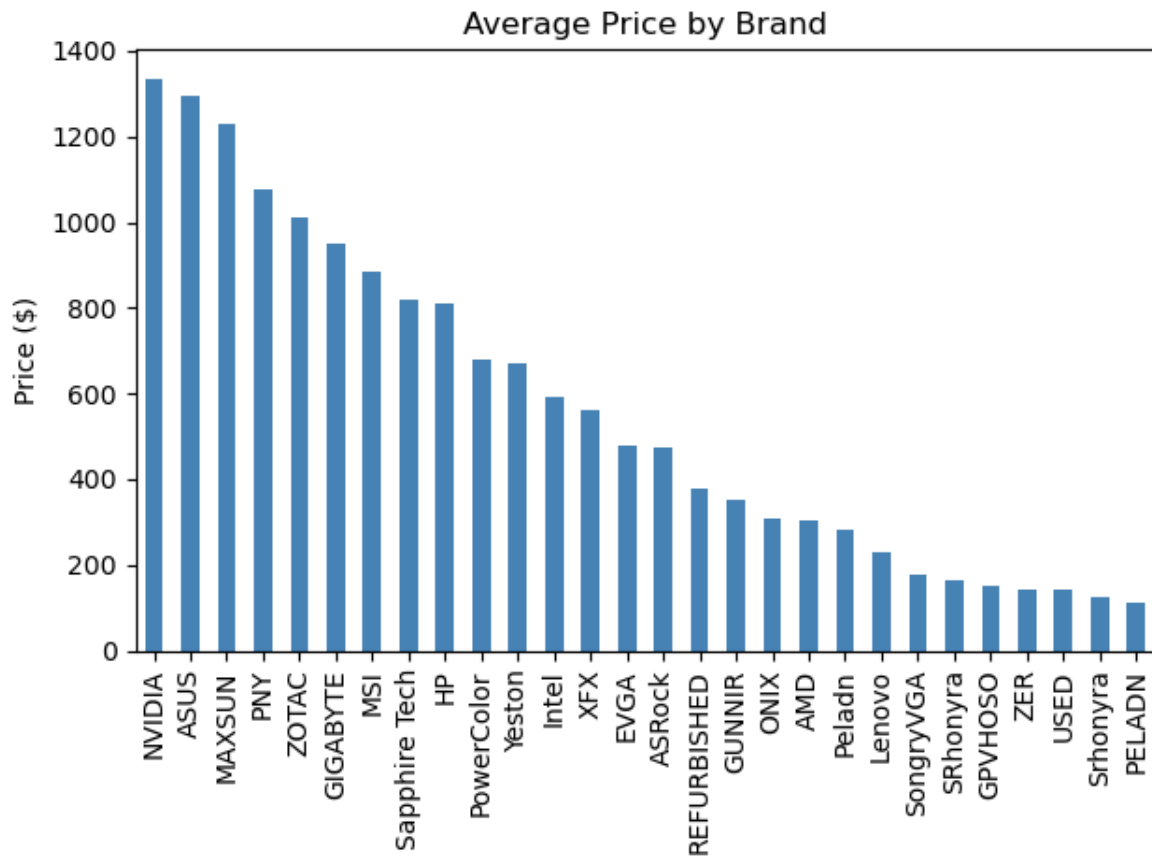
avg_price_brand = df.groupby("brand")["price"].mean().sort_values(ascending=False)
print(avg_price_brand.round(2))

avg_price_brand.plot(kind="bar", color="steelblue")
plt.title("Average Price by Brand"); plt.ylabel("Price ($)"); plt.xlabel("")
plt.tight_layout(); plt.show()

```

brand	
NVIDIA	1336.07
ASUS	1295.50
MAXSUN	1229.20
PNY	1075.94
ZOTAC	1010.33
GIGABYTE	948.74
MSI	885.45
Sapphire Tech	817.39
HP	809.00
PowerColor	677.26
Yeston	669.06
Intel	590.00
XFX	560.76
EVGA	477.52
ASRock	474.20
REFURBISHED	376.66
GUNNIR	353.00
ONIX	309.99
AMD	304.32
Peladn	281.33
Lenovo	230.26
SongryVGA	176.35
SRhonyra	162.22
GPVHOSO	149.99
ZER	143.32
USED	139.99
Srhonyra	125.59
PELADN	113.50

Name: price, dtype: float64



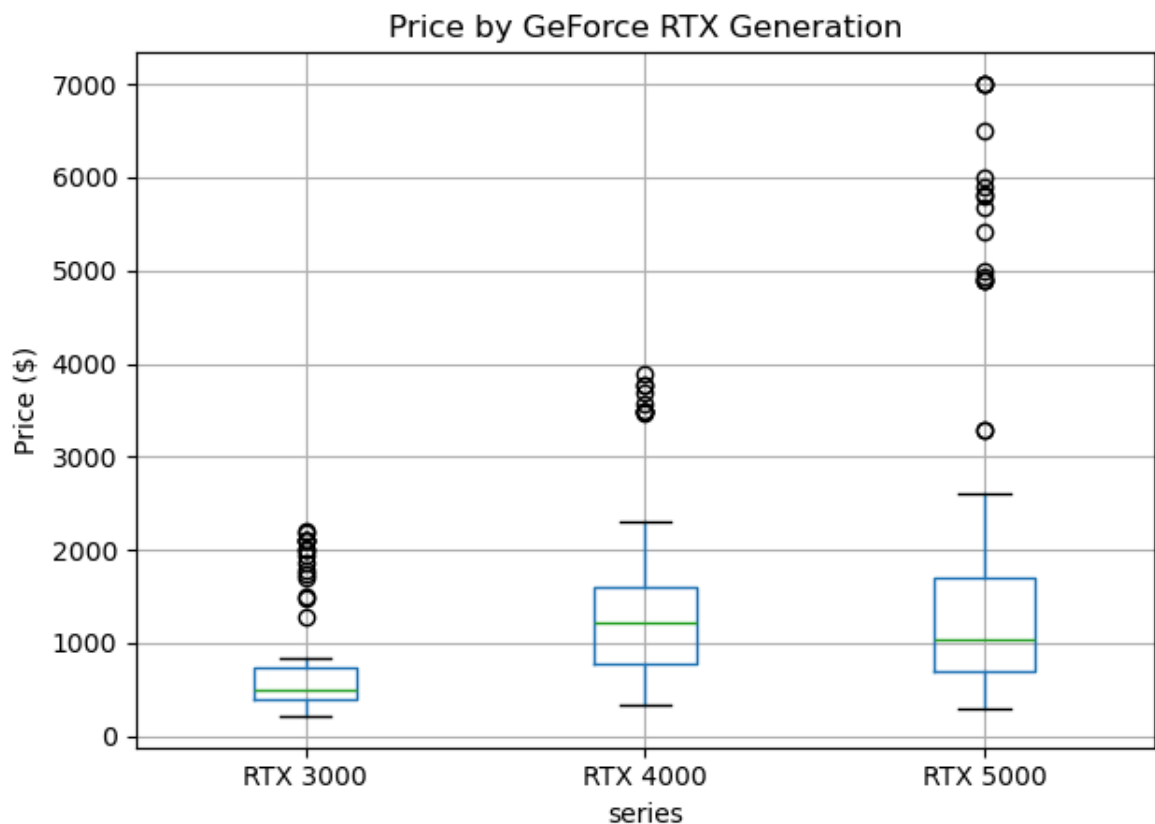
```
In [14]: print(df.groupby("series")["price"].agg(["count", "mean", "min", "max"]).round(2))
print(df.groupby("series")["price_per_gb"].mean().sort_values().round(2))
```

	count	mean	min	max
series				
GTX 1000	38	249.43	97.95	399.00
GTX 1600	21	274.84	179.99	420.00
Other	158	351.58	18.74	2849.00
RTX 3000	90	736.97	219.99	2199.00
RTX 4000	83	1403.01	339.99	3895.00
RTX 5000	176	1514.62	299.99	6999.00
RTX Ada (Pro)	9	3006.11	829.00	8199.00
RX 6000	40	430.92	159.99	789.00
RX 7000	31	1002.15	319.99	1659.99
RX 9000	71	742.60	359.99	1799.99

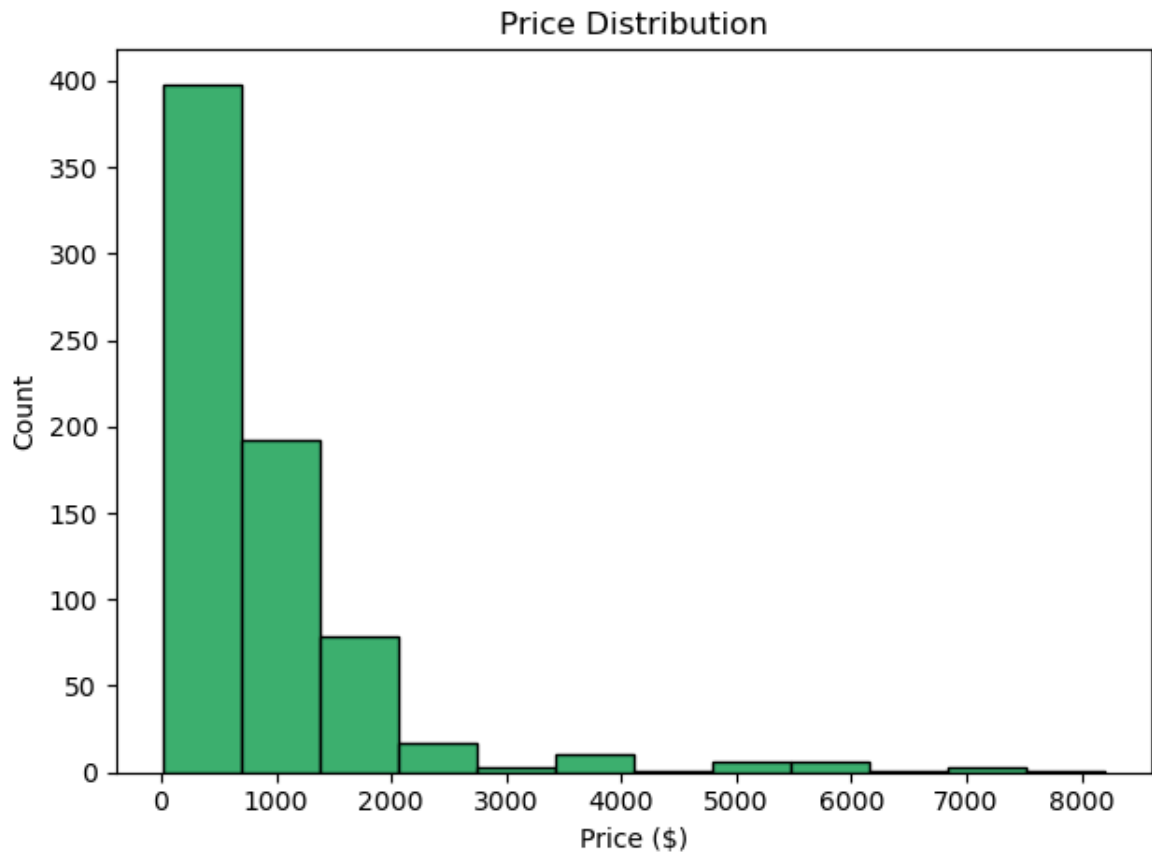
series	
GTX 1000	46.41
RX 6000	49.75
RX 9000	50.31
RX 7000	52.43
Other	54.59
RTX 3000	59.26
GTX 1600	60.96
RTX 4000	90.17
RTX 5000	90.42
RTX Ada (Pro)	108.78

Name: price_per_gb, dtype: float64

```
In [15]: rtx = df[df["series"].isin(["RTX 3000", "RTX 4000", "RTX 5000"])]
rtx.boxplot(column="price", by="series")
plt.title("Price by GeForce RTX Generation"); plt.suptitle("")
plt.ylabel("Price ($)"); plt.tight_layout(); plt.show()
```



```
In [16]: plt.hist(df["price"], bins=12, color="mediumseagreen", edgecolor="black")
plt.title("Price Distribution"); plt.xlabel("Price ($)"); plt.ylabel("Count")
plt.tight_layout(); plt.show()
```



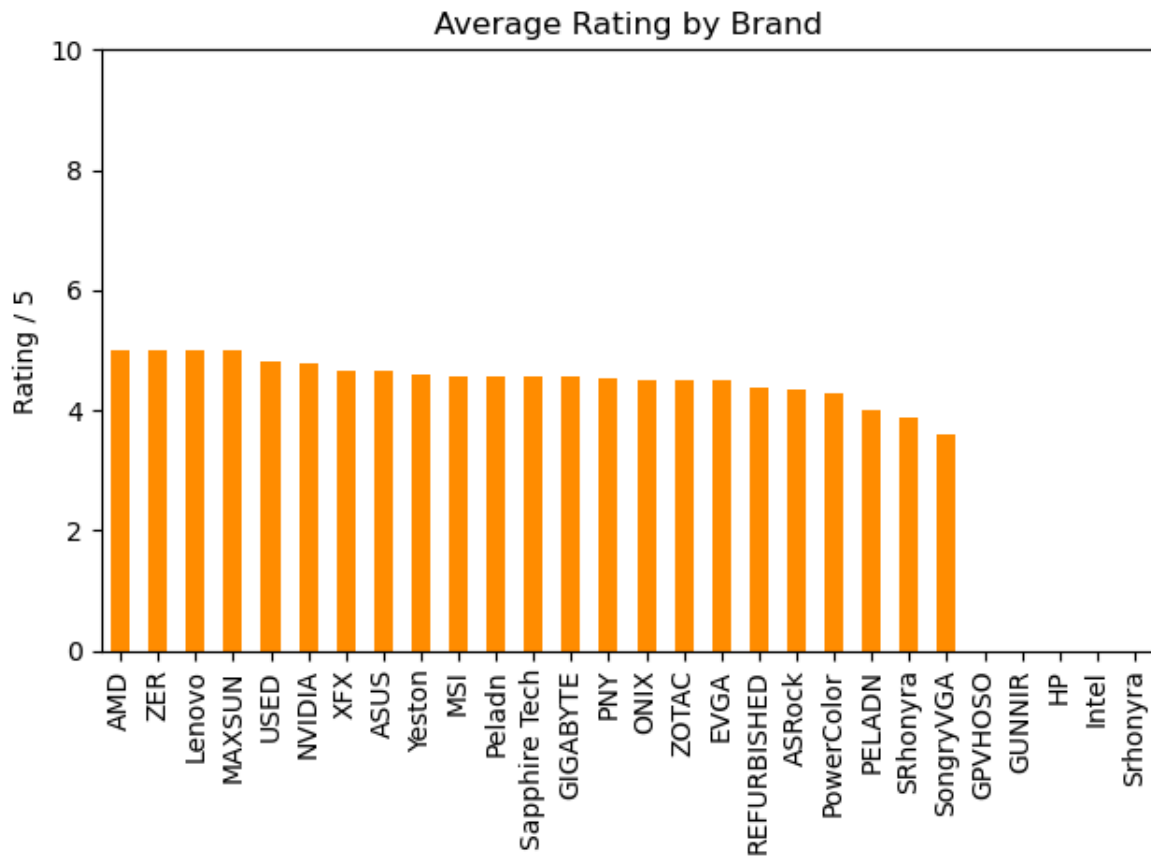
```
In [17]: market_share = df["brand"].value_counts()
print((market_share / len(df) * 100).round(1))
```

```
brand
ASUS                20.2
GIGABYTE            18.4
MSI                 15.6
NVIDIA              6.6
Sapphire Tech       4.7
ZOTAC               4.6
ASRock              4.0
XFX                 3.6
PNY                 3.2
EVGA                2.9
PowerColor          2.9
SongryVGA           2.4
SRhonyra            2.2
Yeston              2.2
Peladn              0.8
AMD                 0.8
REFURBISHED         0.8
Lenovo              0.7
MAXSUN              0.7
GUNNIR              0.7
ZER                 0.4
PELADN              0.3
HP                  0.3
GPVHOSO             0.1
Srhonyra            0.1
ONIX                0.1
Intel               0.1
USED                0.1
Name: count, dtype: float64
```

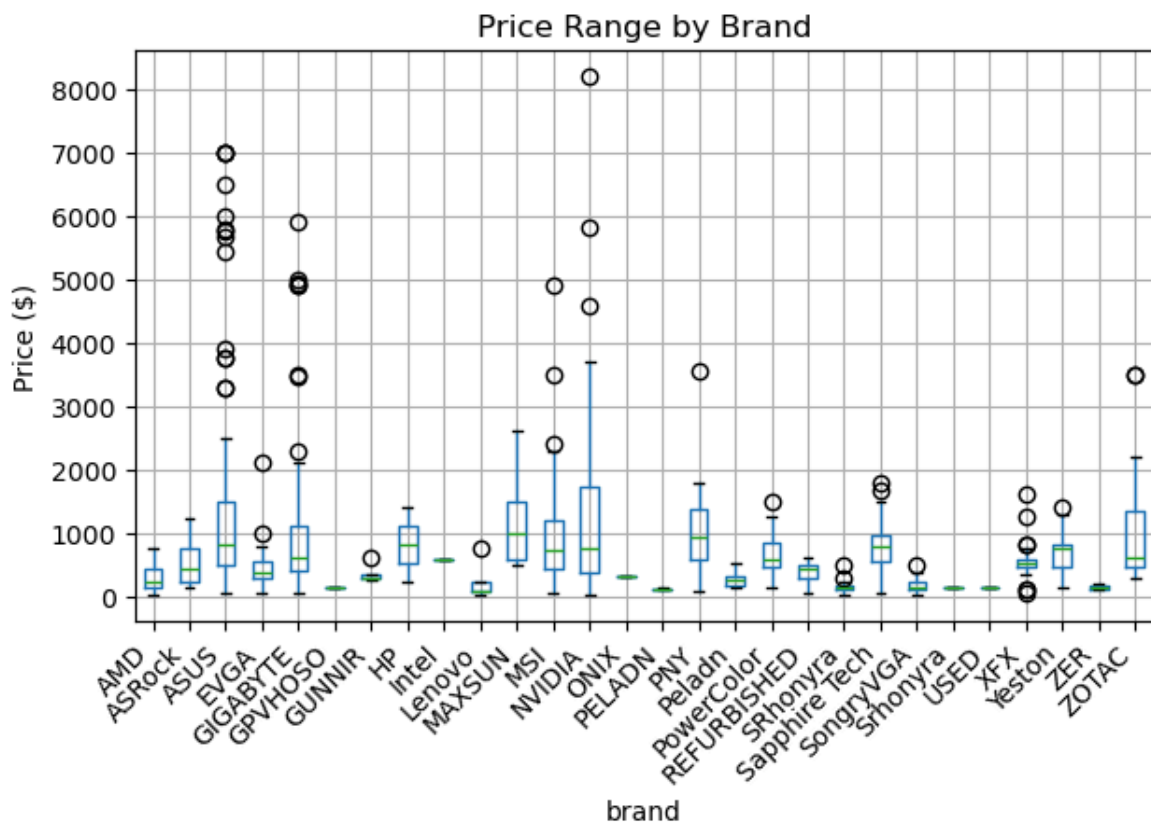
```
In [18]: avg_rating_brand = df.groupby("brand")["rating"].mean().sort_values(ascending=False)
print(avg_rating_brand.round(2))

avg_rating_brand.plot(kind="bar", color="darkorange")
plt.title("Average Rating by Brand"); plt.ylabel("Rating / 5"); plt.ylim(0, 10)
plt.xlabel(""); plt.tight_layout(); plt.show()
```

```
brand
AMD          5.00
ZER          5.00
Lenovo       5.00
MAXSUN       5.00
USED        4.80
NVIDIA      4.77
XFX         4.65
ASUS        4.64
Yeston      4.59
MSI         4.57
Peladn      4.56
Sapphire Tech 4.56
GIGABYTE    4.55
PNY         4.52
ONIX        4.50
ZOTAC       4.49
EVGA        4.49
REFURBISHED 4.38
ASRock      4.35
PowerColor  4.28
PELADN      4.00
SRhonyra    3.88
SongryVGA   3.60
GPVHOSO     NaN
GUNNIR      NaN
HP          NaN
Intel       NaN
Srhonyra    NaN
Name: rating, dtype: float64
```



```
In [19]: df.boxplot(column="price", by="brand")
plt.title("Price Range by Brand"); plt.suptitle("")
plt.ylabel("Price ($)"); plt.xticks(rotation=45, ha="right")
plt.tight_layout(); plt.show()
```



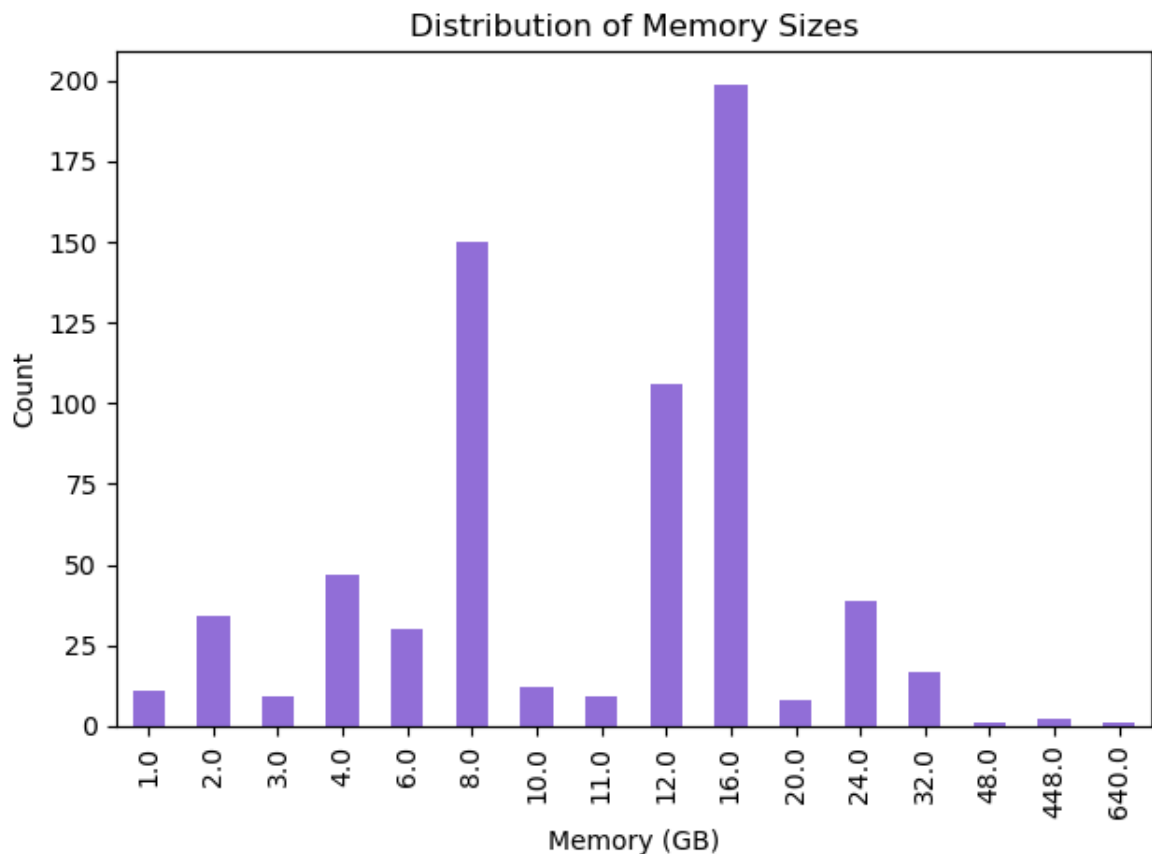
```
In [20]: mem_counts = df["memory_gb"].value_counts().sort_index()
print(mem_counts)
```

```
mem_counts.plot(kind="bar", color="mediumpurple")
plt.title("Distribution of Memory Sizes"); plt.xlabel("Memory (GB)"); plt.ylabel
plt.tight_layout(); plt.show()
```

memory_gb

1.0	11
2.0	34
3.0	9
4.0	47
6.0	30
8.0	150
10.0	12
11.0	9
12.0	106
16.0	199
20.0	8
24.0	39
32.0	17
48.0	1
448.0	2
640.0	1

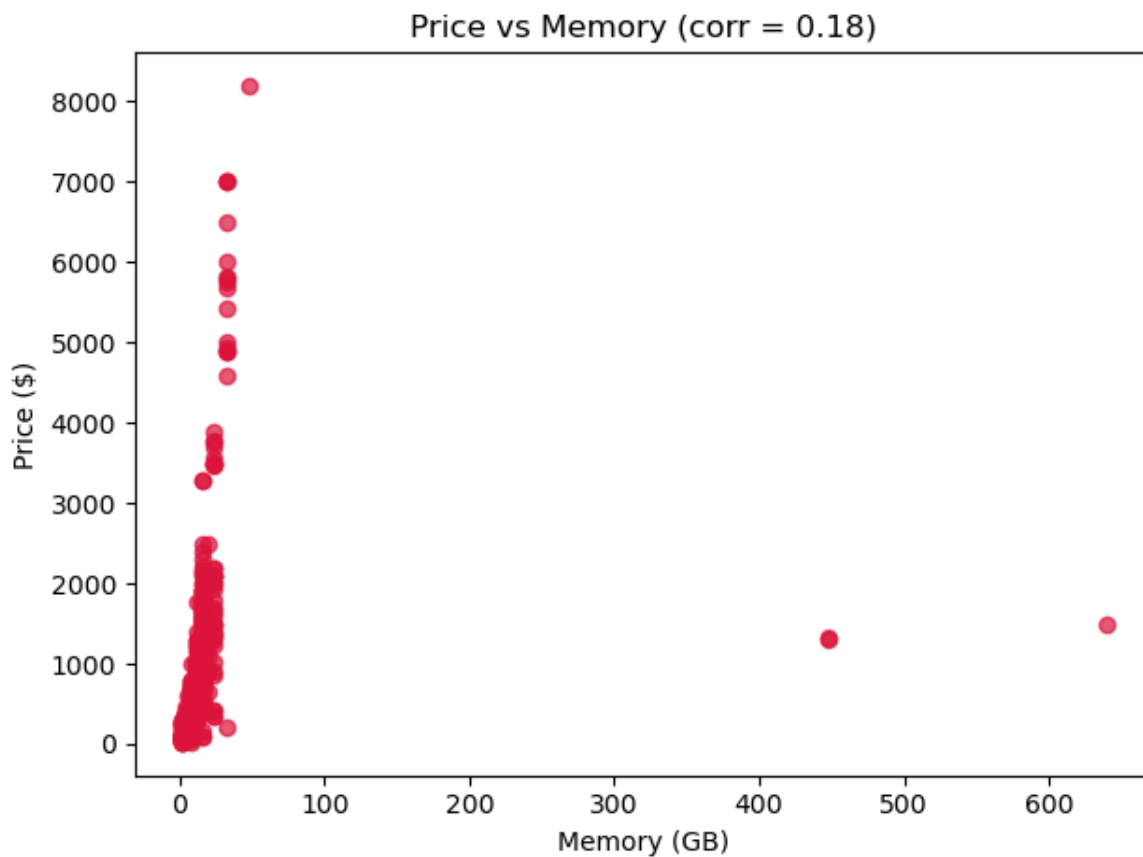
Name: count, dtype: int64



```
In [21]: corr = df["memory_gb"].corr(df["price"])
print("correlation:", round(corr, 2))

plt.scatter(df["memory_gb"], df["price"], alpha=0.7, color="crimson")
plt.title("Price vs Memory (corr = %.2f)" % corr)
plt.xlabel("Memory (GB)"); plt.ylabel("Price ($)")
plt.tight_layout(); plt.show()
```

correlation: 0.18



```
In [22]: df.groupby("brand")["memory_gb"].agg(lambda x: x.mode()[0])
```

```
Out[22]: brand
AMD          12.0
ASRock       16.0
ASUS         16.0
EVGA         8.0
GIGABYTE     8.0
GPVHOSO      6.0
GUNNIR       8.0
HP           2.0
Intel        12.0
Lenovo       1.0
MAXSUN       8.0
MSI          16.0
NVIDIA       16.0
ONIX         12.0
PELADN       4.0
PNY          16.0
Peladn       8.0
PowerColor   16.0
REFURBISHED  12.0
SRhonyra     2.0
Sapphire Tech 16.0
SongryVGA    4.0
Srhonyra     3.0
USED         3.0
XFX          16.0
Yeston       16.0
ZER          4.0
ZOTAC        8.0
Name: memory_gb, dtype: float64
```

```
In [23]: df["segment"] = pd.cut(df["price"], bins=[0, 500, 1200, float("inf")],
                                labels=["Entry-level (<$500)", "Mid-range ($500-1200)", "
                                High-end (>$1200)"])

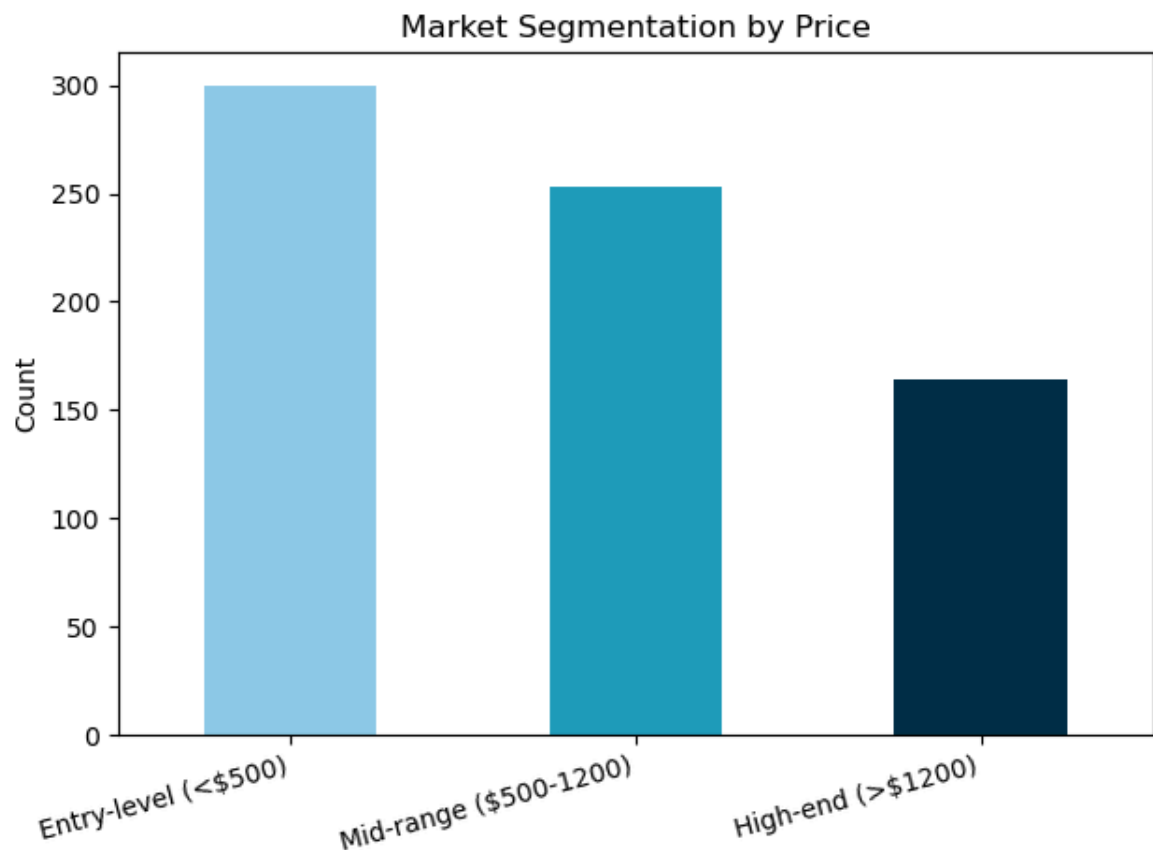
seg_table = df.groupby("segment").agg(
    count=("price", "size"), avg_price=("price", "mean"),
    avg_memory=("memory_gb", "mean"), avg_rating=("rating", "mean")).round(2)
print(seg_table)

seg_table["count"].plot(kind="bar", color=["#8ecae6", "#219ebc", "#023047"])
plt.title("Market Segmentation by Price"); plt.ylabel("Count"); plt.xlabel("")
plt.xticks(rotation=15, ha="right"); plt.tight_layout(); plt.show()
```

C:\Users\Lenovo\AppData\Local\Temp\ipykernel_18192\865701589.py:4: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
seg_table = df.groupby("segment").agg(
    count="size", avg_price="price", avg_memory="memory_gb", avg_rating="rating")
```

segment	count	avg_price	avg_memory	avg_rating
Entry-level (<\$500)	300	289.34	7.22	4.43
Mid-range (\$500-1200)	253	767.25	12.63	4.60
High-end (>\$1200)	164	2239.74	29.12	4.66



```
In [24]: df.to_csv("newegg_gpus_clean.csv", index=False)
print("Saved newegg_gpus_clean.csv")
```

Saved newegg_gpus_clean.csv

In []: